Desarrollo de una herramienta en Python para el postprocesado de modelos térmicos realizados con ESATAN-TMS

Trabajo de Fin de Grado
ESPECIALIDAD: Propulsión Aeroespacial

Autor: Asier Fiel Mariblanca

Tutor: David González Bárcena

Madrid, junio 2026

Contents

1. Abstract **1**

2. Introduction **3**

2.1. Objectives 3

3. State of Art **5**

3.1. Thermal Tools in the Aerospace Industry 5

3.2. Post-Processing Capabilities in ESATAN-TMS 5

3.3. Custom Interfaces: JPL Example 5

3.4. Practical Application: FalconSat-2 5

3.5. Motivation for the Present Work 5

4. Theoretical Background **7**

4.1. Thermal Control in Engineering 7

4.2. Space Thermal Environment 7

4.3. Heat Transfer Mechanisms 7

4.3.1. Conduction Heat Transfer 7

4.3.2. Radiation Heat Transfer 7

4.4. Mathematical Modeling of Thermal Systems 7

4.4.1. Disadvantages of the Finite Element Method (FEM) 7

4.4.2. Lumped Parameter Method (LPM) 7

4.5. Software and Advanced Methods for Thermal Simulation 9

4.6. Analytical Capabilities of ESATAN-TMS 9

5. Development of the Thermal Model in ESATAN-TMS and Data Export/Analysis 11
5.1. Subsystem Classification . . . 11
5.2. Materials and Properties . . . 11
5.2.1. Thermo-optical properties . . . 11
5.3. Steady-state radiative case . . . 11
5.4. Boundary Conditions and Loads Definition . . . 11
5.5. Model Verification: Group-Level Energy Balances . . . 11
5.6. Data provided by the TMD . . . 11
5.6.1. Use in the Python post-processing . . . 11

6. Design and Implementation of the Post-Processing Code 13
6.1. Code Structure and Architecture . . . 13
6.1.1. Motivation for a Modular Architecture . . . 13
6.1.2. Files overview . . . 13
6.2. Software Stack and Dependencies . . . 13
6.3. Filtering and Grouping Methods . . . 13
6.4. Figure Conventions and Representation Criteria . . . 13
6.5. Scope of Examples and Cryogenic-Flow Validation . . . 13

7. Results and Visualisations from the Post-Processing Workflow 15
7.1. Filtering Results . . . 15
7.2. Sankey Diagram . . . 15
7.3. Network Graph Views (NX) . . . 15
7.4. Stacked-Bar Balances . . . 15
7.5. Cryogenic-Flow Views (ARIEL) . . . 15

8. Conclusions and Future Work 17
8.1. Conclusions . . . 17
8.2. Future Work . . . 17

References 19

Anexos 21

List of Figures

4.1. Simplified heat transfer system and its corresponding thermal network [1] 8

List of Tables

Acronyms

ESA,European Space Agency

,

Capítulo 1

Abstract

This Bachelor's Thesis focuses on

Capítulo 2

Introduction

2.1. Objectives

Capítulo 3

State of Art

3.1. Thermal Tools in the Aerospace Industry

3.2. Post-Processing Capabilities in ESATAN-TMS

3.3. Custom Interfaces: JPL Example

3.4. Practical Application: FalconSat-2

3.5. Motivation for the Present Work

Capítulo 4

Theoretical Background

4.1. Thermal Control in Engineering

In the field of engineering, temperature control is essential to keep systems and components within safe thermal limits. This is especially critical in aerospace systems where extreme variations can negatively impact structural safety and operational reliability.
Therefore, extending systems' durability and improving their performance are crucial key factors [2].

4.2. Space Thermal Environment

4.3. Heat Transfer Mechanisms

4.3.1. Conduction Heat Transfer

4.3.2. Radiation Heat Transfer

4.4. Mathematical Modeling of Thermal Systems

4.4.1. Disadvantages of the Finite Element Method (FEM)

4.4.2. Lumped Parameter Method (LPM)

The Lumped Parameter Method (LPM) is the standard mathematical approach for spacecraft thermal analysis.
This method is a discretization technique that simplifies a continuous physical system into a finite number of discrete elements called nodes. The fundamental assumption is that each node is isothermal, meaning its temperature is uniform throughout its entire volume. This allows the system to

be solved as a network of interconnected thermal resistances and capacitances, analogous to an electrical circuit, where temperature equates to voltage and heat flow to current flow. [3]

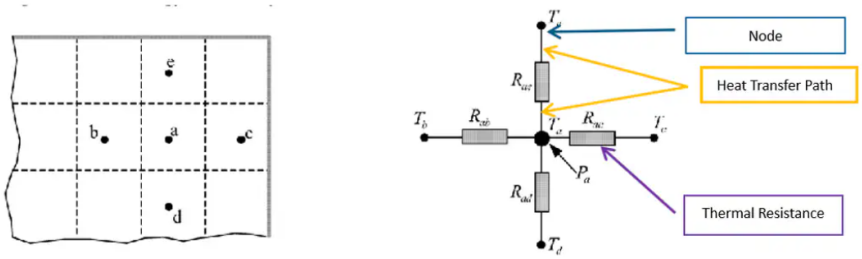

Figura 4.1: Simplified heat transfer system and its corresponding thermal network [1]

The behaviour of the network can be described as a system of differential equations whose coefficients are grouped values of the nodes. The mathematical expression for the nodal energy balance equation is:

$$C_i \frac{dT_i}{dt} = Q_i - \sum_{j=1}^{n} GL_{ij}(T_i - T_j) - \sum_{j=1}^{n} GR_{ij}(T_i^4 - T_j^4) \tag{4.1}$$

where:

- C_i [J/K]: thermal capacitance of node i ($m_i\ c_{p,i}$); stores/releases heat as T_i changes.
- T_i [K]: temperature of node i.
- t [s]: time.
- GL_{ij} [W/K]: conductive conductance between nodes i and j. The value is zero if there is not a conductive path.
- GR_{ij} [W/K^4]: radiative coupling factor between i and j in the non-linear T^4 form; it embeds σ, view factors and emissivities for diffuse–gray surfaces.
- Q_i [W]: net heat source applied to node i (internal dissipation, heaters, and absorbed external fluxes as modelled in the network).
- $\sum_{j=1}^{n}$: sum over all nodes j, which are thermally connected to i. The terms $GL_{ij}(T_i - T_j)$ and $GR_{ij}(T_i^4 - T_j^4)$ contribute as inflow when the expression in parentheses is negative.

The discrete approximation has advantages in numerical as well as experimental work, and lends itself to simple hand calculations as well as to computerised solution. A continuous system obeys a differential equation, whereas a networked model obeys a system of algebraic equations, when studying a steady-state case, in a finite number of variables which can be solved by standard techniques. [3]
Therefore, it is especially useful in aerospace applications, such as the thermal modeling of satellites, where a balance between accuracy and efficiency is required . [2]

4.5. Software and Advanced Methods for Thermal Simulation

4.6. Analytical Capabilities of ESATAN-TMS

Capítulo 5

Development of the Thermal Model in ESATAN-TMS and Data Export/Analysis

5.1. Subsystem Classification

5.2. Materials and Properties

5.2.1. Thermo-optical properties

5.3. Steady-state radiative case

5.4. Boundary Conditions and Loads Definition

5.5. Model Verification: Group-Level Energy Balances

5.6. Data provided by the TMD

5.6.1. Use in the Python post-processing

Capítulo 6

Design and Implementation of the Post-Processing Code

6.1. Code Structure and Architecture

6.1.1. Motivation for a Modular Architecture

6.1.2. Files overview

6.2. Software Stack and Dependencies

6.3. Filtering and Grouping Methods

6.4. Figure Conventions and Representation Criteria

6.5. Scope of Examples and Cryogenic-Flow Validation

Capítulo 7

Results and Visualisations from the Post-Processing Workflow

7.1. Filtering Results

7.2. Sankey Diagram

7.3. Network Graph Views (NX)

7.4. Stacked-Bar Balances

7.5. Cryogenic-Flow Views (ARIEL)

Capítulo 8

Conclusions and Future Work

8.1. Conclusions

8.2. Future Work

References

[1] Understanding the lumped-parameter thermal network method [online].

[2] C. A. Carballo, Desarrollo de una herramienta en python para el postprocesado de modelos térmicos procedentes de esatan-tms, Tech. rep., Universidad Politécnica de Madrid, Escuela Técnica Superior de Ingeniería Aeronáutica y del Espacio (2025).

[3] ESATAN-TMS Thermal Engineering Manual, ITP Engines UK Ltd, Whetstone, Leicester, UK, August 2015.

Título del anexo

Aquí puedes meter la información que no sea imprescindible en el cuerpo del trabajo pero si que interese que esté en el documento.